Market Share Analysis: Android vs iOS

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Graph 1: Market Share Over Time - $x(t)$ and $y(t)$

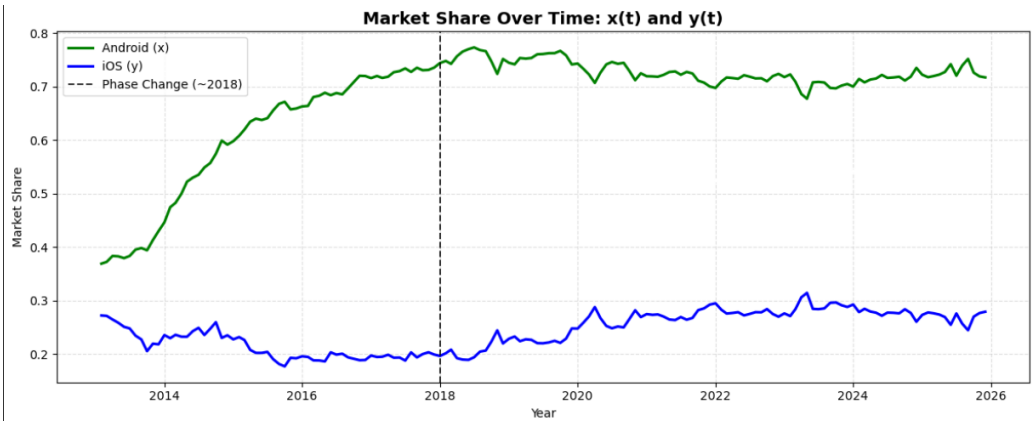

Figure 1: Market share over time: $x(t)$ (Android) and $y(t)$ (iOS)

Observation:

Android share grew rapidly from 0.37 in 2013 to approximately 0.75 by 2018, after which growth slowed and the curve flattened around 0.70 - 0.75. iOS share declined from 0.27 in 2013 to a low of 0.19 around 2016, then slowly recovered to approximately 0.28 and remained stable.

Conclusion:

Growth is clearly nonlinear. Android shows fast early growth followed by saturation. iOS shows a decline followed by stabilization. Neither curve is linear or exponential(both flatten over time). This behaviour is consistent with a logistic growth model.

Graph 2: Total Share $x(t) + y(t)$

Observation:

Total share started at approximately 0.64 in 2013 and rose steadily, reaching a late-period mean of 0.995 by 2020 onwards. The curve approaches 1.0 asymptotically and flattens completely after 2020.

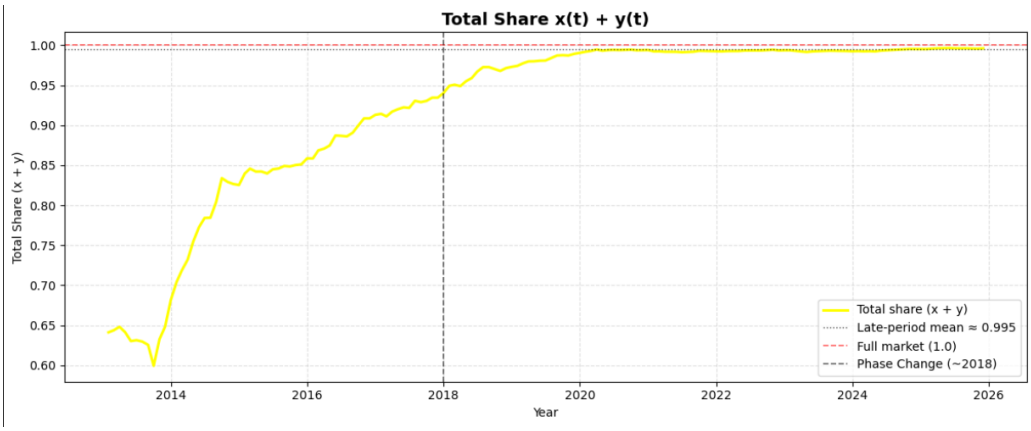

Figure 2: Total combined market share $x(t) + y(t)$ over time

Conclusion:

Saturation exists. The total market share converges to 1.0, i.e. Android and iOS together cover virtually the entire smartphone OS market. This confirms that the market has a finite carrying capacity K. The assumption of using $\frac{x+y}{K}$ in the Lotka–Volterra model is fully valid.

Graph 3: Derivatives $\dot{x}(t)$ and $\dot{y}(t)$

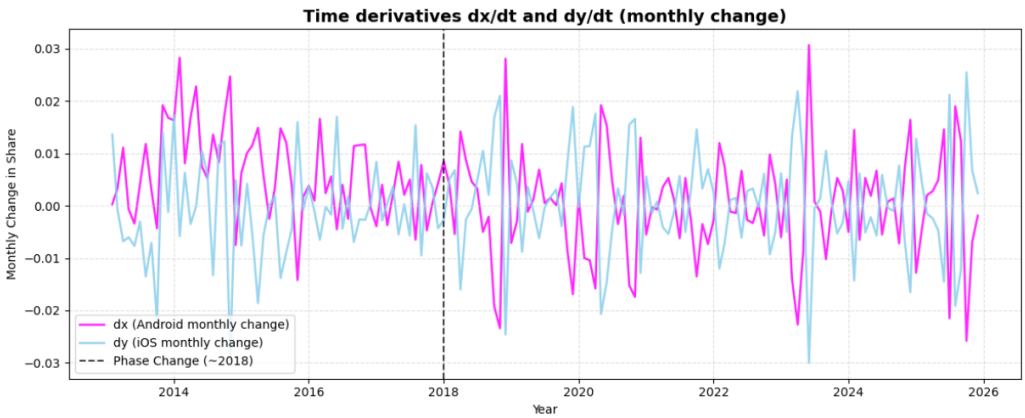

Figure 3: Time derivatives dx/dt and dy/dt over time

Observation:

Both derivatives show large fluctuations in the early period (2013 - 2016), with spikes reaching ± 0.025. Over time the magnitude of these fluctuations decreases. In the later period (2020 - 2025) both dx and dy stay mostly within ± 0.015. The sign of dx shifts from mostly positive before 2018 to balanced around zero after 2018.

Conclusion:

Large changes early and small changes later confirm that the system is slowing down and moving toward stability. The change in sign of dx shows Android stopped its net growth

trend. This is characteristic of nonlinear dynamics. A phase change is visible around 2018.

Graph 4: Rolling Volatility and Phase Portrait

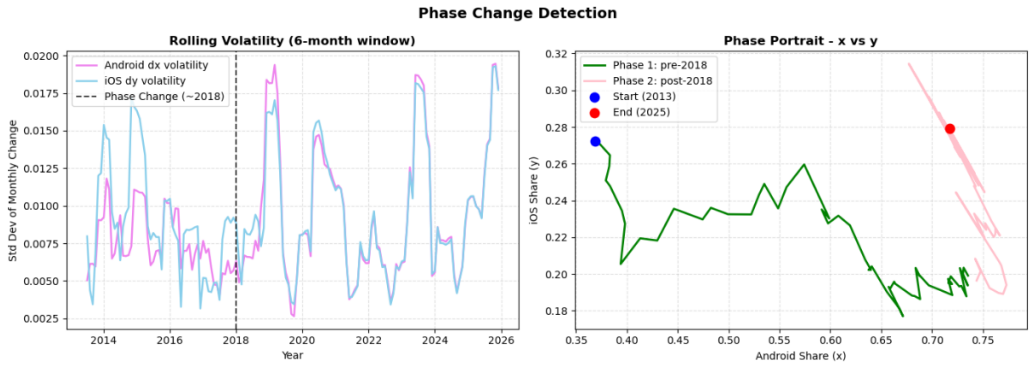

Figure 4: Rolling volatility and phase portrait of $x(t)$ vs $y(t)$

Observation:
The rolling volatility plot shows high volatility in 2013-2016 and generally lower volatility after 2018, with irregular spikes. The phase portrait clearly shows two distinct trajectory regimes:

- A long green path (pre-2018) representing fast movement across a wide range of x and y values.
- A tight pink cluster (post-2018) representing slow fluctuations near equilibrium.

Detected Phase Change: approximately 2018.

- **Phase 1 (pre-2018):** Fast growth, active competition, large derivatives, high volatility. Android rapidly gaining market share while iOS declining.
- **Phase 2 (post-2018):** Slow movement, near equilibrium, small derivatives, low volatility. Both shares stabilized.

Conclusion:
The system has two clearly distinct phases. The behaviour before and after 2018 is fundamentally different. A single set of parameters cannot describe both phases, which directly justifies piecewise parameter fitting.